Pre-Calculus — Cheat Sheet

Procadamia | Key Formulas & Rules

FACTORING QUICK REFERENCE

Pattern	Formula
Diff of Squares	$a^2-b^2 = (a+b)(a-b)$
Sum of Cubes	$a^3+b^3 = (a+b)(a^2-ab+b^2)$
Diff of Cubes	$a^3-b^3 = (a-b)(a^2+ab+b^2)$
Trinomial (a=1)	$x^2+bx+c = (x+p)(x+q)$ *where $p \cdot q=c$, $p+q=b$*
Trinomial (a≠1)	*AC method: find factors of ac that sum to b*
Perfect Square	$a^2 \pm 2ab+b^2 = (a \pm b)^2$

Always factor out GCF first!

QUADRATIC FORMULA & COMPLEX NUMBERS

Quadratic Formula

$$x = (-b \pm \sqrt{(b^2-4ac)}) / 2a$$

Discriminant	$\Delta = b^2-4ac$ *$\Delta>0$: 2 real | $\Delta=0$: 1 real | $\Delta<0$: 2 complex*

Complex Numbers (i)

$i = \sqrt{(-1)}$	$i^2 = -1$
$i^3 = -i$	$i^4 = 1$
Cycle repeats every 4: i, −1, −i, 1	
Conjugate	$(a+bi)(a-bi) = a^2+b^2$

FUNCTIONS: DOMAIN, RANGE & INVERSES

Concept	Rule
Domain restrictions	*√: radicand ≥ 0 | fraction: denom ≠ 0 | log: arg > 0*
Even function	$f(-x) = f(x) \rightarrow$ y-axis symmetry
Odd function	$f(-x) = -f(x) \rightarrow$ origin symmetry
Find inverse	*Swap x&y, solve for y → $f^{-1}(x)$*
Composite	$f(g(x))$ — plug g into f
Discontinuity	*Removable (hole) vs. Non-removable (asymptote)*

HLT: passes → function has an inverse

POLYNOMIALS: END BEHAVIOR & ZEROS

Leading Term	Left (x→−∞)	Right (x→+∞)
$+x^n$ (even)	↑ Up	↑ Up
$-x^n$ (even)	↓ Down	↓ Down
$+x^n$ (odd)	↓ Down	↑ Up
$-x^n$ (odd)	↑ Up	↓ Down

Zero Multiplicity

Odd mult.	*Crosses x-axis*
Even mult.	*Touches (bounces off) x-axis*

Degree = max # of zeros; # turns ≤ degree−1

SYNTHETIC DIVISION & RATIONAL ZEROS

- **Synthetic Division:** Divide poly by (x−c); if remainder = 0, c is a zero
- **Remainder Theorem:** f(c) = remainder when f(x) ÷ (x−c)
- **Factor Theorem:** (x−c) is a factor ↔ f(c) = 0

Possible Rational Zeros (PRZ)

PRZ = ±(factors of constant) / (factors of leading

Sign graph: plot zeros on number line, test intervals for +/−

LOGARITHMS

Rule	Formula
Definition	$\log_b(x) = y \leftrightarrow b^y = x$
Product	$\log(xy) = \log(x)+\log(y)$
Quotient	$\log(x/y) = \log(x)-\log(y)$
Power	$\log(x^n) = n \cdot \log(x)$
Change of Base	$\log_b(x) = \ln(x)/\ln(b)$
Natural log	$\ln(e^x) = x$; e^(ln x) = x
Common values	*log(1)=0; log(10)=1; ln(1)=0; ln(e)=1*

To solve: isolate log, convert to exponential; or log both sides

TRIGONOMETRY QUICK REFERENCE

SOH-CAH-TOA

$\sin\theta$	*= opp/hyp*
$\cos\theta$	*= adj/hyp*
$\tan\theta$	*= opp/adj*
$\csc\theta$	*= 1/sin*
$\sec\theta$	*= 1/cos*
$\cot\theta$	*= 1/tan*

Angle	sin	cos	tan
0°	0	1	0
30°	1/2	√3/2	1/√3
45°	√2/2	√2/2	1
60°	√3/2	1/2	√3
90°	1	0	undef

Key Identities

$\sin^2\theta + \cos^2\theta = 1$	
$1 + \tan^2\theta = \sec^2\theta$	
$1 + \cot^2\theta = \csc^2\theta$	
Law of Sines	$a/\sin A = b/\sin B = c/\sin C$
Law of Cosines	$c^2 = a^2+b^2-2ab \cdot \cos C$
Radians	$\theta(\text{rad}) = \theta(°) \times \pi/180$

ASTC (All Students Take Calculus): Quadrant I → All +; II → Sin +; III → Tan +; IV → Cos +

PIECEWISE FUNCTIONS

- Evaluate using the rule whose domain condition matches x
- **Continuity:** pieces must connect at boundaries
- **Closed dot (•):** point included; **Open dot (○):** excluded
- Check limits from left & right at each break point

EXAM TIPS

- Factor before canceling — never cancel unfactored terms
- State domain restrictions before simplifying rational expressions
- Verify inverse: $f(f^{-1}(x)) = x$ AND $f^{-1}(f(x)) = x$
- PRZ ≠ actual zeros — must verify with synthetic division
- Expand log of products; condense sums of logs back to one log
- Radians vs degrees — know which mode your calculator is in!